Jasper Sands

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RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for quantum-device characterization and control.

EDUCATION

Columbia University, New York, NY — Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning · Machine Learning for Functional Genomics.

Washington University in St. Louis, St. Louis, MO — Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

Coursework: Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison.

TECHNICAL REPORTS

Kate Bonner and Jasper Sands. *Noise-Aware Ansatz Design for Digitized Adiabatic Quantum Computing.* 2025.

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* 2025.

John C. Papciak and Jasper Sands. *Multimodal Hyperbolic Embeddings for AMR Detection and Taxonomic Modeling in Metagenomic Contigs.* 2025.

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* — Sep 2026 – May 2027

- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, targeting manual tuning and interpretation of charge measurements that bottleneck semiconductor quantum-hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Benchmarking automated classification against expert manual interpretation of the same device measurements.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* — Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and optimizing precision under a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* — May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling with Dr. Venkat Venkatasubramanian by grounding model outputs in ontology and molecular-graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher* — Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents, with human-in-the-loop feedback for cross-state policy analysis.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* · MIT iQuHACK 2026

- Wrote GPU-accelerated state preparation, threshold oracle, and bisection search for Iterative Quantum Amplitude Estimation; measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly*

- Built a browser-native symplectic stabilizer simulator for rotated and XZZX surface codes with real-time syndrome visualization and MWPM/Union-Find decoding; finite-size scaling estimated thresholds of 3.2% and 2.7% respectively.

Q-Search: proof-gated quantum algorithm research and verification engine

- Automated search for structural quantum advantage on non-abelian hidden subgroup, dihedral coset, and linear code equivalence problems, pairing frontier theoretical reasoning with agentic engineering.
- Formal proof gates test each proposed mechanism against 1,200+ classical baselines across 720+ theorem modules, with negative results permanently indexed so the engine does not re-propose a dead end.

Noise-Aware Adiabatic Simulation: transverse-field Ising model · *Qiskit* · IBM Heron r2

- Digitized adiabatic ground-state preparation on IBM Heron r2 using its parameterised RZZ(θ) pulses rather than fixed two-qubit decompositions, with palindromic Trotter steps for second-order accuracy.
- Flattened the annealing schedule near the critical point to slow evolution as the spectral gap closes.

TEACHING EXPERIENCE

Introduction to Quantum Computing, Columbia University · *Teaching Assistant* Jan 2026 – May 2026

Parallel and Concurrent Programming, Washington University · *Teaching Assistant* Jan 2025 – May 2025

Reverse Engineering and Malware Analysis, Washington University · *Teaching Assistant* Aug 2024 – Dec 2024

Taught a lecture in the quantum computing and malware analysis courses, plus office hours and grading.

INDUSTRY EXPERIENCE

Smack Technologies · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulations for core AI pipelines, integrating knowledge-graph state representations to expand synthetic corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency, raising throughput 64×.
- Validated simulator outputs against independently selected reference cases, bounding error to 0.55%, in an automated report that regenerates as the physics model changes.

Highnote · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, with 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern* May 2023 – Jul 2023

- Built internal admin and secure-query tooling with Retool, JavaScript, and Ruby on Rails, replacing direct production database writes and saving roughly 250 engineering hours/month.

COMPETITIONS

- **Overall Winner** & **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street/Classiq Challenge, qualified for the NYU Abu Dhabi Quantum and AI Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Hackathon 2026.

TECHNICAL SKILLS

Quantum tools: Qiskit, Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP.

Quantum algorithms: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT, block encoding, linear combination of unitaries, quantum walks, VQE, QAOA, Trotter decomposition.

Quantum error correction: Rotated and XZZX surface codes, stabilizer formalism, MWPM and Union-Find decoding, finite-size scaling, threshold estimation, noise modeling.

Programming: Python, C++, Rust, C, MATLAB, JavaScript/TypeScript, Go, Java.

ML & systems: PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy, Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group, Sep 2025 – Jun 2026. Led weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms notes, paired with simulation checks.
- **Founding team member**, Columbia Autonomous Racing Club. Coordinated sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.
- **Team Captain**, WashU Gymnastics Club.